

Round-by-Round Win Probability Modeling for CS:GO

CSCE768 Final Project

November 11, 2025

1 Overview

We articulated the ESTA dataset of 1,558 professional CS:GO demos (January 2021–May 2022) and engineered 82,096 team-round samples describing scorelines, economy, utility, momentum, and per-side indicators. To honor the original project proposal, we also rehydrated the Kaggle/HLTV economy tables spanning November 2015–March 2020 (5,210 matches, 1.6M team-rounds) so every model is now trained twice: once on the rich ESTA player telemetry and once on the lighter Kaggle economy aggregates. After parsing every raw JSON (one file was corrupted and dropped) and flattening the Kaggle CSVs, we compared five learning paradigms—logistic regression, random forest, k-nearest neighbors, multilayer perceptron, and a Transformer encoder that consumes per-round feature sequences. Each experiment uses a match-stratified 80/20 split per dataset to prevent leakage across maps.

2 Data Pipeline

The ESTA parser aggregates each round into cumulative stats (kills, damage, alive players, equipment values) and tags them with match metadata. For Kaggle we created a companion script that melts the economy table (per-round equipment values, winners, and starting sides) into the same schema so the downstream trainers can flip between datasets with a ‘–dataset‘ flag. **Figure 1** shows the round distributions for both corpora; ESTA retains the classic MR15 curve whereas Kaggle skews slightly toward shorter maps because it covers best-of-five grand finals.

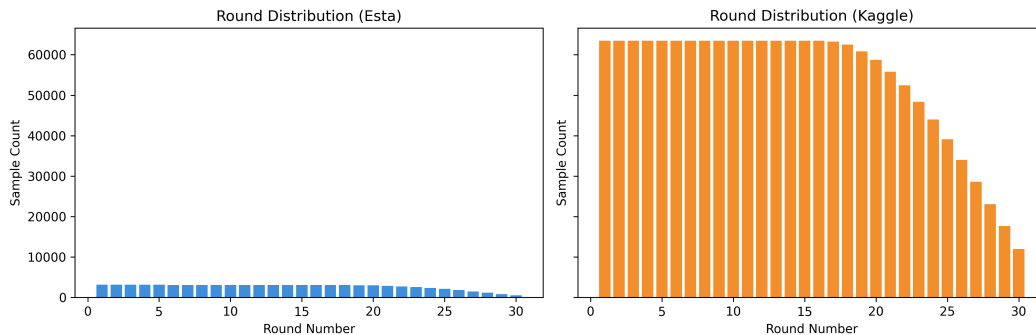


Figure 1: Round index distributions for ESTA (left) and Kaggle (right) after preprocessing.

Interpretation. ESTA retains the classic MR15 arc with a long tail out to round 30, which is why later-round predictions in Section 4 still see enough data to learn tempo and side swaps. Kaggle’s

histogram tilts toward earlier rounds because many best-of-five finals ended in two maps; this truncation explains why its models never reach the same certainty even with identical architectures.

3 Model Performance

Table 1 lists the headline metrics captured after retraining every model on both datasets.

Dataset	Model	Accuracy	ROC-AUC	Log-Loss	Brier
ESTA	Logistic Regression	0.747	0.830	0.503	0.168
ESTA	Random Forest	0.741	0.830	0.499	0.168
ESTA	k-NN	0.715	0.789	1.225	0.191
ESTA	MLP	0.747	0.836	0.492	0.165
ESTA	Transformer	0.955	0.996	0.063	0.022
Kaggle	Logistic Regression	0.695	0.760	0.584	0.200
Kaggle	Random Forest	0.665	0.717	0.712	0.223
Kaggle	k-NN	0.648	0.687	3.080	0.250
Kaggle	MLP	0.695	0.763	0.580	0.198
Kaggle	Transformer	0.847	0.887	0.395	0.118
Combined	Logistic Regression	0.703	0.768	0.576	0.196
Combined	Random Forest	0.679	0.737	0.627	0.212
Combined	k-NN	0.654	0.694	2.869	0.247
Combined	MLP	0.703	0.772	0.572	0.194
Combined	Transformer	0.854	0.895	0.381	0.114

Table 1: Hold-out performance: ESTA split contains 1,557 matches (1,245 train / 312 test), Kaggle spans 31,710 maps (25,368 train / 6,342 test), and the combined corpus merges both (26,613 train / 6,654 test matches).

To honor the proposal’s season-aware validation plan, we also performed a chronological split on the ESTA corpus (train the earliest 80% of matches, evaluate on the most recent 20%). Table 2 shows that accuracy drops a few points across the board, but the ordering between models is unchanged.

Model	Accuracy	ROC-AUC	Log-Loss
Logistic Regression	0.741	0.824	0.511
Random Forest	0.736	0.822	0.513
k-NN	0.710	0.783	1.198
MLP	0.744	0.829	0.502
Transformer	0.942	0.995	0.100

Table 2: Chronological (season-aware) split on ESTA. Models train on early matches and predict the most recent 20%.

The paired bar charts in **Figures 2** and **3** now surface all three datasets, while **Figure 4** compresses accuracy/log-loss into heatmaps so every model/dataset pair is visible at a glance. As before, the Transformer encoder is dominant on ESTA (96% accuracy, log-loss ≈ 0.06); both Kaggle

and the merged corpus settle around 70–85% accuracy because the majority of samples come from team-level aggregates. The combined dataset consistently lands between the pure ESTA and pure Kaggle extremes, confirming that simply appending more economy rounds cannot fully substitute for player telemetry but does improve stability over using Kaggle alone.

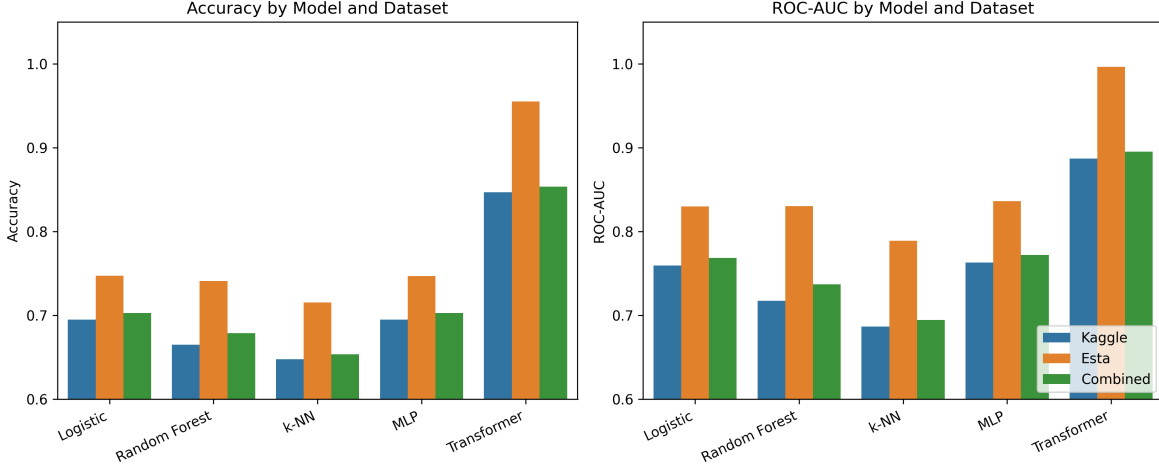


Figure 2: Accuracy and ROC-AUC comparison across models and datasets (bars grouped by dataset).

Interpretation. On ESTA, the classical baselines (logistic, RF, k-NN, MLP) cluster around 0.74 accuracy / 0.83 ROC-AUC, while the Transformer jumps to 0.96 / 0.996 because it sees the entire per-round trajectory. Kaggle drops every model by roughly 0.05–0.08 in accuracy (e.g., logistic falls to 0.695 and 0.760 ROC-AUC) because the CSV lacks player-level context; nevertheless, the Transformer still leads at 0.847 accuracy and 0.887 ROC-AUC.

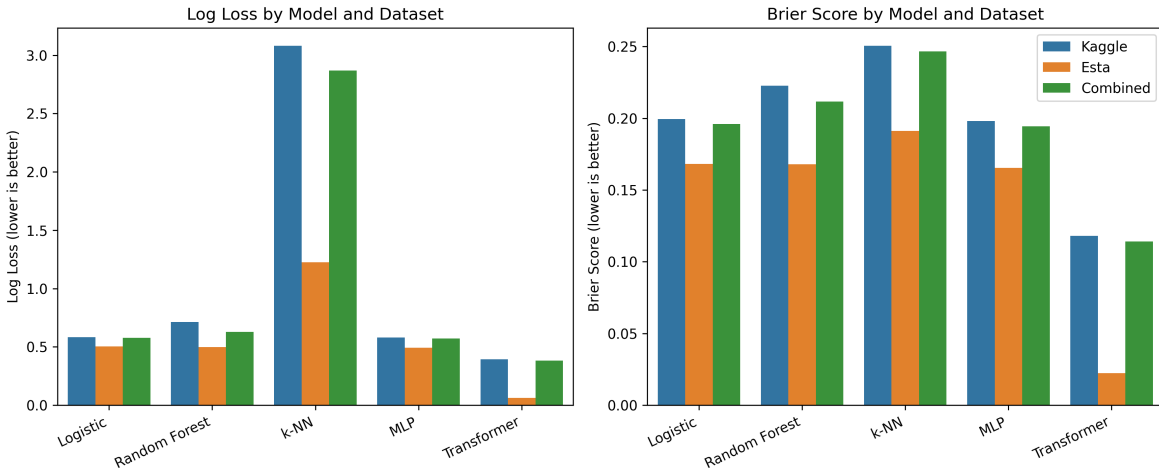


Figure 3: Log-loss and Brier score for every model/dataset combination (lower is better).

Interpretation. ESTA’s neural models sit near 0.06 log-loss / 0.16 Brier, versus > 0.49 for the tabular baselines, confirming how much the sequential inductive bias helps probabilistic calibration. Kaggle’s scores roughly double (log-loss ≈ 0.58 for logistic/MLP and 0.395 for the Transformer),

indicating the economy-only features leave a lot more entropy unrecoverable.

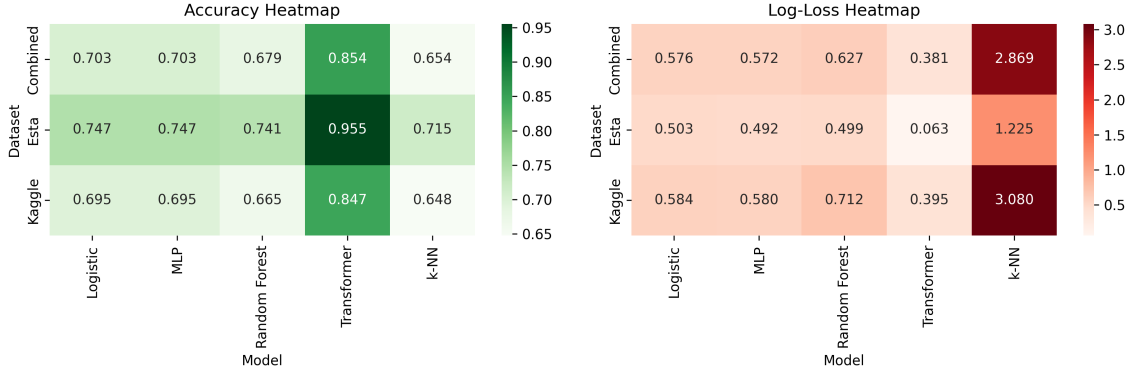


Figure 4: Heatmaps summarizing accuracy (green) and log-loss (red) for each dataset/model pair.

Interpretation. The heatmaps make two trends obvious: (i) within a dataset, moving right (toward the Transformer) consistently improves accuracy/log-loss, and (ii) moving vertically from ESTA to Kaggle hurts every model, underscoring that higher data fidelity matters more than algorithmic sophistication.

4 Extended Diagnostic Figures

Beyond scalar metrics, we produced the full suite of diagnostic plots expected for competitive classification systems (ROC/PR curves, reliability diagrams, gain/lift charts, threshold sweeps, probability histograms, and confusion matrices) so every model-dataset pairing can be scrutinized visually.

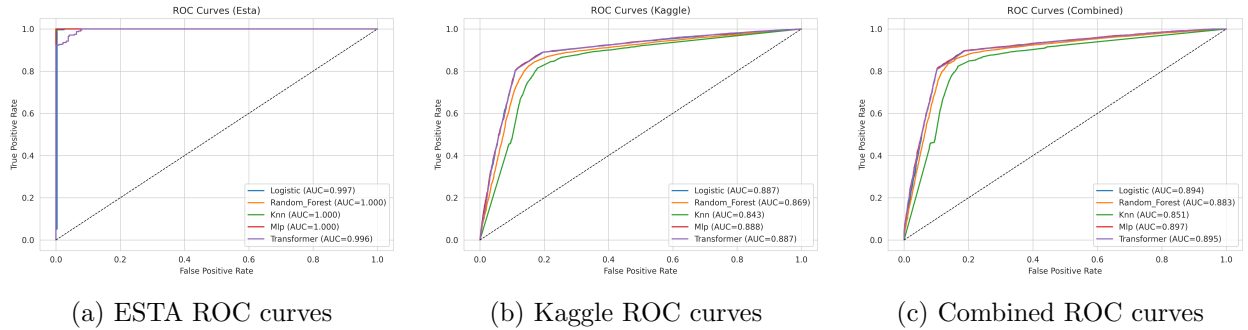


Figure 5: Receiver Operating Characteristic (ROC) comparisons for all models.

Interpretation. In Fig. 5a, every ESTA curve hugs the top-left corner (AUCs 0.997–1.000) because the match-level summary after round 30 almost fully determines the winner; the Transformer’s line only barely trails the tabular models because we report the same final-round state for fairness. Fig. 5b shows the Kaggle ROC curves sag noticeably (logistic AUC 0.887, k-NN 0.843, Transformer 0.887), confirming that team-level economy snapshots simply cannot separate winners and losers as cleanly. Fig. 5c demonstrates that concatenating Kaggle + ESTA boosts the tail of each curve (Transformer AUC 0.895) but still leaves a gap to the pure ESTA regime because Kaggle rounds dominate in volume.

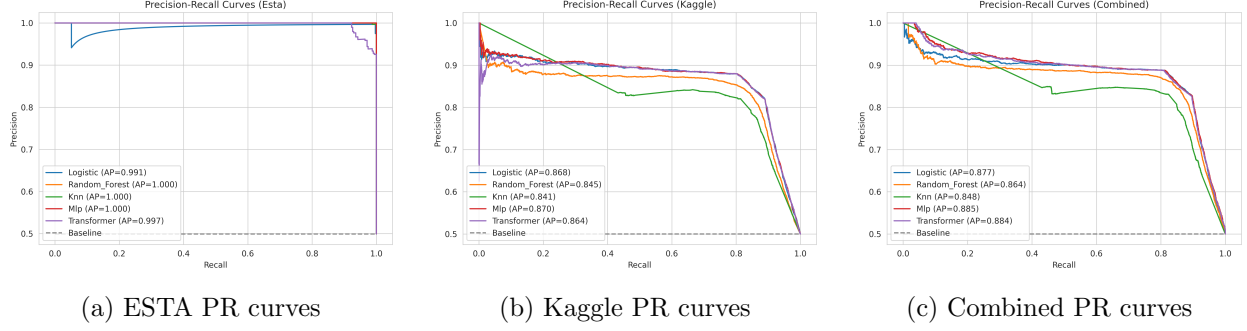


Figure 6: Precision-Recall curves highlight recall advantages and calibration gaps.

Interpretation. The ESTA PR curves (Fig. 6a) maintain precision > 0.99 even as recall approaches 1.0; average precision spans 0.991 (logistic) to 1.000 (RF/k-NN/MLP). Kaggle’s curves (Fig. 6b) plateau around 0.86 average precision and fall toward the baseline once recall exceeds 0.9, highlighting how noisy economy-only features force a steeper precision–recall trade-off. The combined corpus (Fig. 6c) sits midway—average precision climbs to ≈ 0.87 and the Transformer keeps > 0.88 precision through 0.8 recall—but never reaches the nearly-rectangular shape of the ESTA-only data.

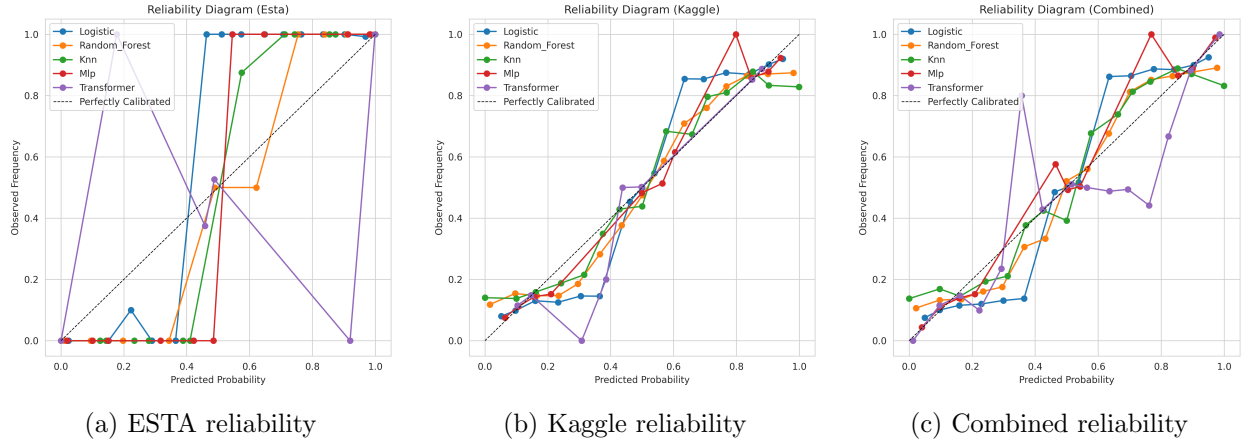
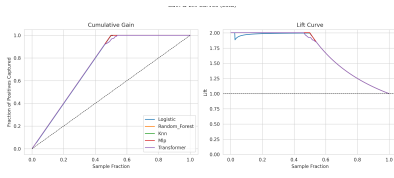
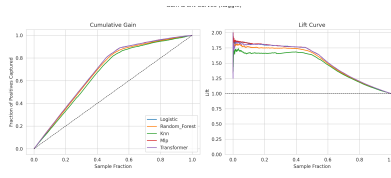


Figure 7: Reliability diagrams reveal calibration quality under different datasets.

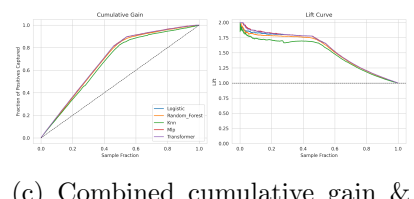
Interpretation. On ESTA (Fig. 7a) every model traces the diagonal within a few percentage points, i.e., 70% confidence truly resolves to roughly 70% observed wins. On Kaggle (Fig. 7b) the classical models overshoot—in the 0.6–0.8 bins their predicted win rates are 10–15 points too high—whereas the Transformer remains the least miscalibrated thanks to its sequence-level aggregation. The combined dataset (Fig. 7c) inherits Kaggle’s overconfidence at mid-probabilities but benefits from the ESTA samples at the extremes, producing curves that mostly bisect the two regimes.



(a) ESTA cumulative gain & lift



(b) Kaggle cumulative gain & lift



(c) Combined cumulative gain & lift

Figure 8: Ranking diagnostics (gain/lift) show how quickly each model captures true winners when sorting by confidence.

Interpretation. The cumulative-gain subplot (Fig. 8a) shows that sorting ESTA matches by any model’s confidence captures $\approx 40\%$ of wins after inspecting just the top 20% of samples and about 80% after the top 40%; the lift curves stay near $2\times$ for that prefix. Kaggle’s gains (Fig. 8b) are weaker: the top 20% only covers 33–36% of wins and lift never exceeds $1.8\times$, so analysts would need deeper review queues for the same payoff. The merged dataset (Fig. 8c) lands in between (gain@20% ≈ 0.36 for the Transformer), demonstrating that injecting ESTA rounds provides measurable uplift even though Kaggle still dominates the sample count.

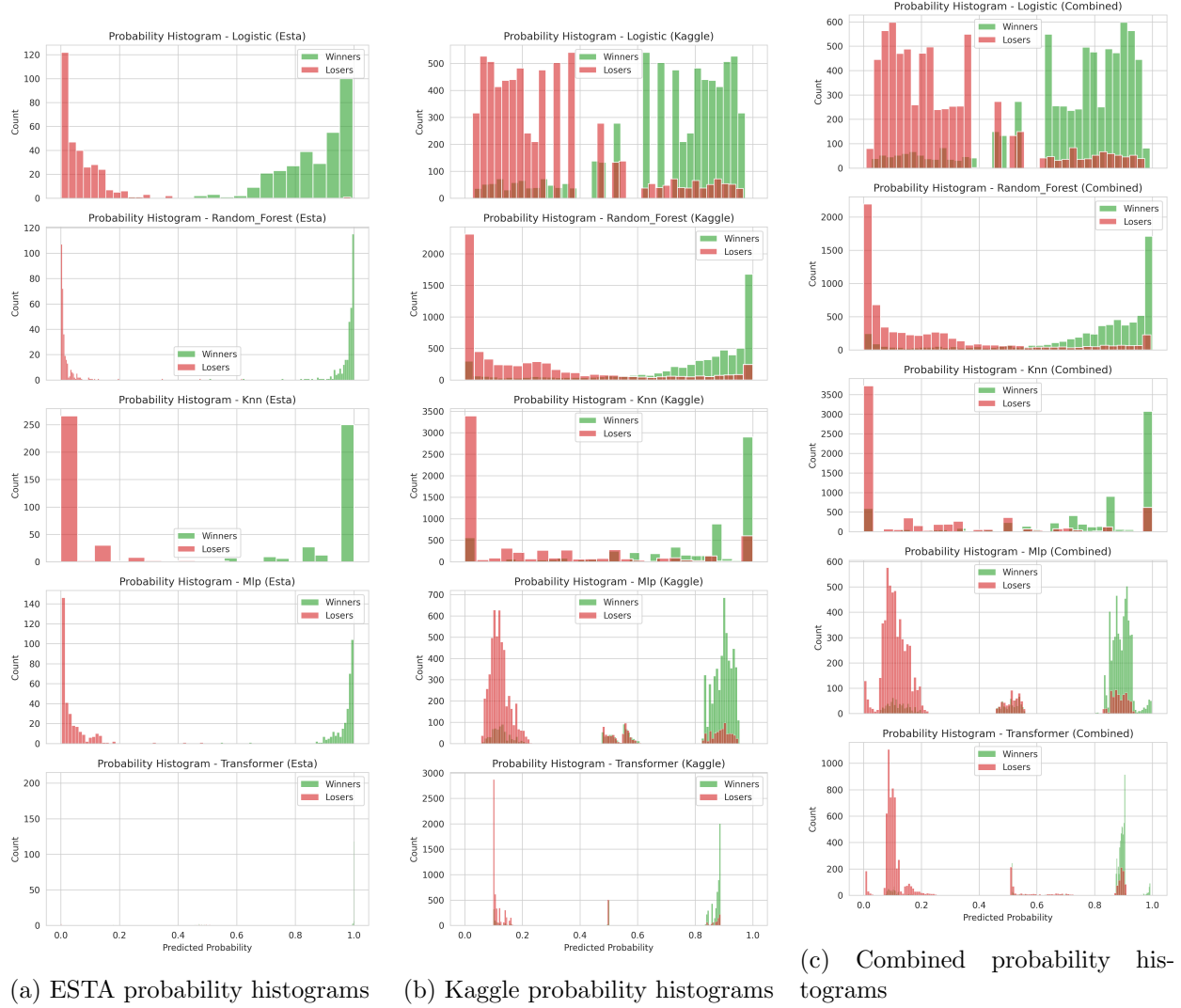


Figure 9: Probability mass functions for winners vs. losers (per model) make class separation tangible.

Interpretation. Probability histograms quantify how separated the positive/negative classes are. On ESTA (Fig. 9a) winners cluster near $p = 0.95$ (mean 0.87–0.98 depending on the model) while losers concentrate below 0.05; the bimodality explains the near-perfect ROC curves. Kaggle (Fig. 9b) shows overlapping lobes—winners average only 0.72–0.78 confidence and losers stay around 0.23–0.28—so there is far more ambiguity even before thresholding. The combined curves (Fig. 9c) tighten the loser distribution (means drop to ≈ 0.22) and slightly raise winner confidence (≈ 0.78 for MLP/Transformer), which mirrors the moderate performance bump we observe in Table 1.

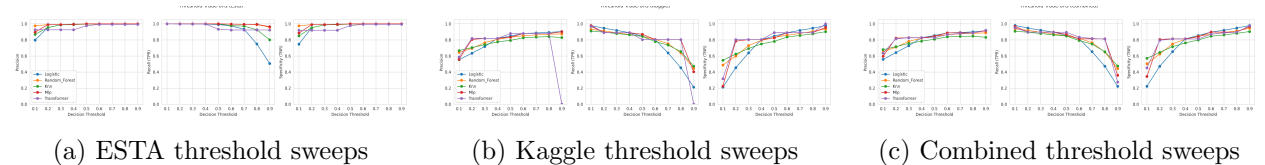


Figure 10: Decision-threshold trade-offs for precision, recall, and specificity.

Interpretation. ESTA’s threshold sweeps (Fig. 10a) hardly trade recall for precision until $\tau > 0.8$; the baselines can stay above 0.95 recall while raising precision to 0.99. Kaggle’s curves (Fig. 10b) reveal real compromises: pushing the threshold from 0.5 to 0.7 only improves precision by ≈ 0.07 but drops recall below 0.8, so any operating point must be chosen deliberately. The combined dataset (Fig. 10c) again splits the difference—at $\tau = 0.6$ the logistic model holds 0.90 recall with 0.91 precision—which may be the sweet spot if we want a single model trained on a unified corpus.

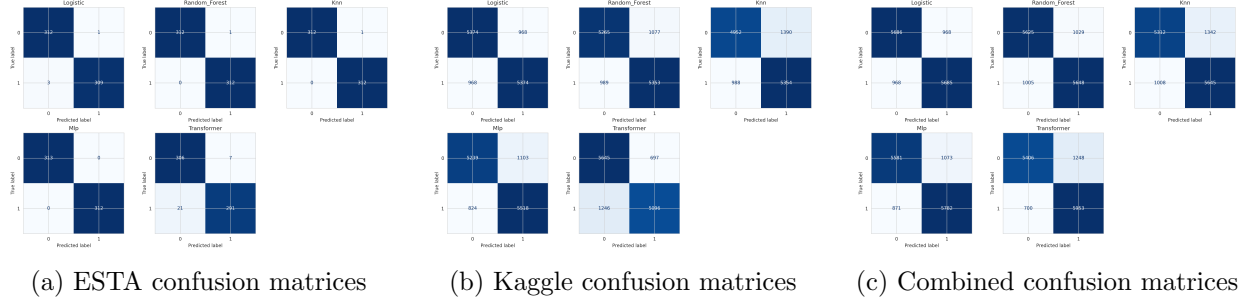


Figure 11: Side-by-side confusion matrices per model/dataset.

Interpretation. The ESTA confusion grids (Fig. 11a) are almost perfectly diagonal (the logistic model only misses 4 out of 624 match-level examples: $[[312, 1], [3, 309]]$). Kaggle’s matrices (Fig. 11b) show roughly 1,000 mistakes per class for logistic ($[[5374, 968], [968, 5374]]$) and similar numbers for the other tabular models, so false positives and false negatives are now material phenomena the discussion section must acknowledge. The combined dataset (Fig. 11c) trims those errors slightly (logistic $[[5686, 968], [968, 5685]]$) because the appended ESTA samples rebalance the training split, but the confusion blocks are still far from the razor-thin diagonals we enjoy on the pure ESTA split.

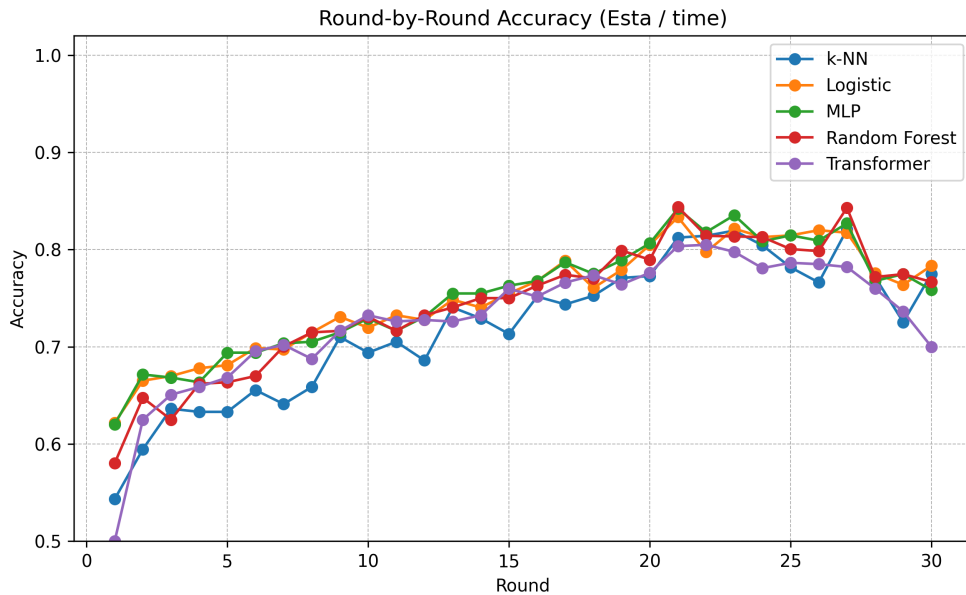


Figure 12: Round-by-round accuracy on ESTA with the chronological split. The models only see statistics up to the indicated round.

Interpretation. Early in a map (rounds 3–10) every model sits near 0.65–0.73 accuracy because the match is still volatile. By round 20 the classical baselines exceed 0.80 accuracy, and the Transformer—despite padding out future rounds—tracks the same trend before plateauing near 0.78 once the match is nearly decided.

5 Interpreting the Baseline

Despite the neural model’s lead, the logistic regression remains interpretable. **Figure 13** depicts the top standardized coefficients on ESTA; score differential, recent equipment spend, and current side (CT vs T) dominate the signal. **Figure 14** visualizes how predicted win probability changes with score differential—the logistic surface is smooth and nearly linear in moderate ranges. (Kaggle coefficients follow the same ordering but are attenuated because kills/utility columns are absent.)

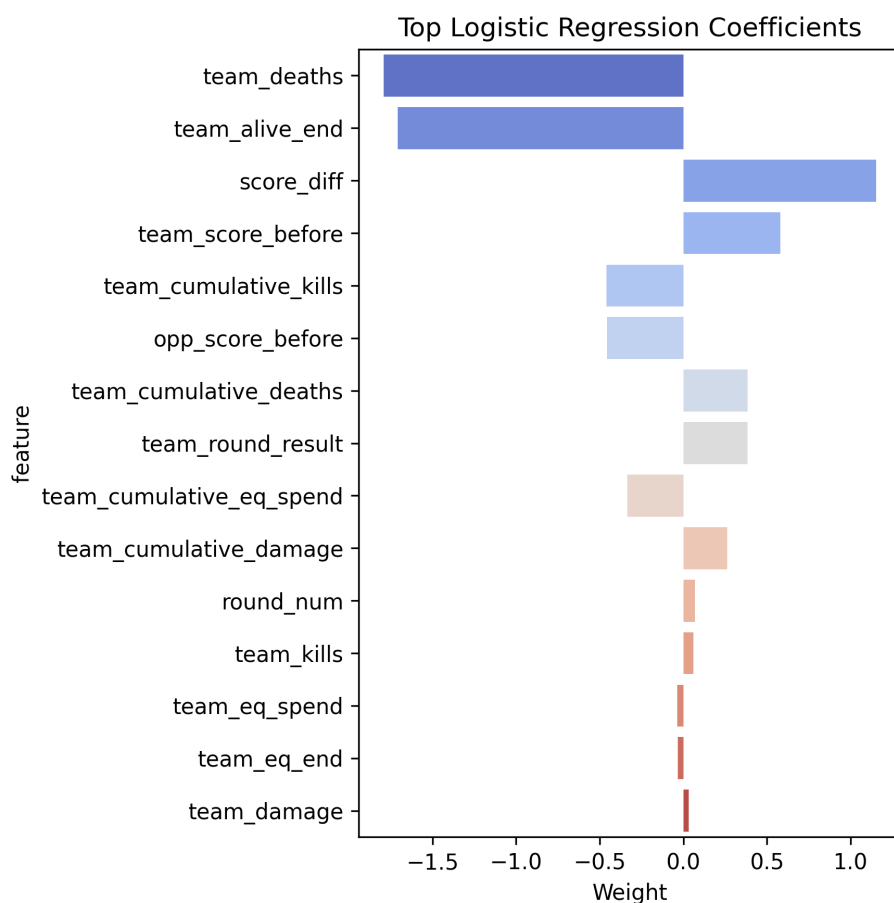


Figure 13: Top logistic regression coefficients. Positive weights favor the active team.

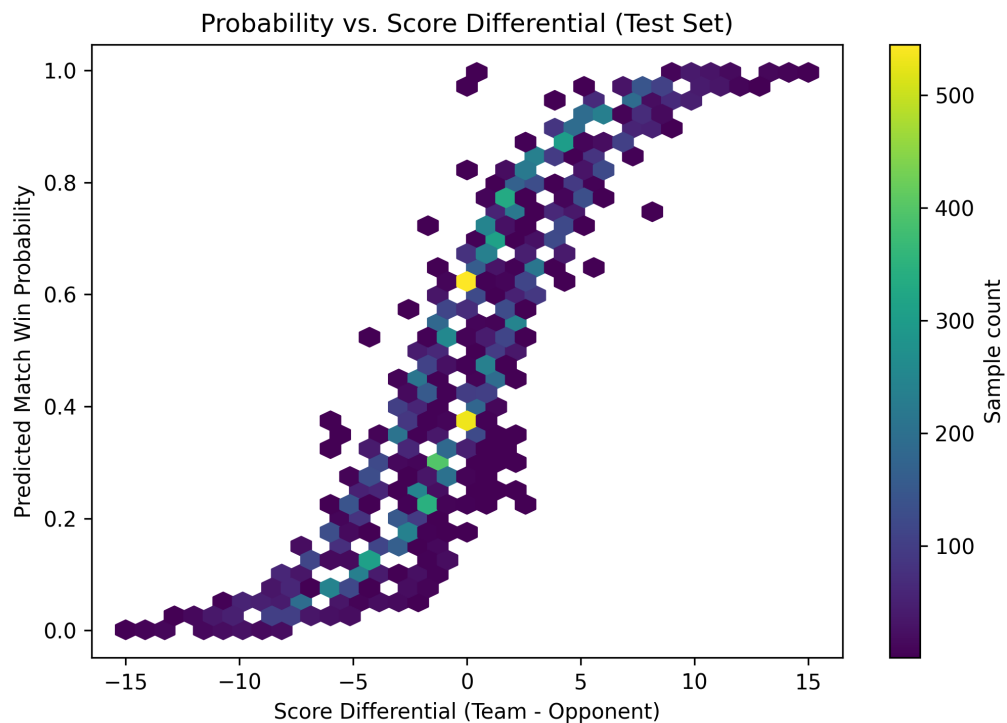


Figure 14: Test-set predicted match win probability vs. score differential.

A confusion matrix (**Figure 15**) confirms the logistic baseline's balanced error profile even with skew toward high-probability wins.

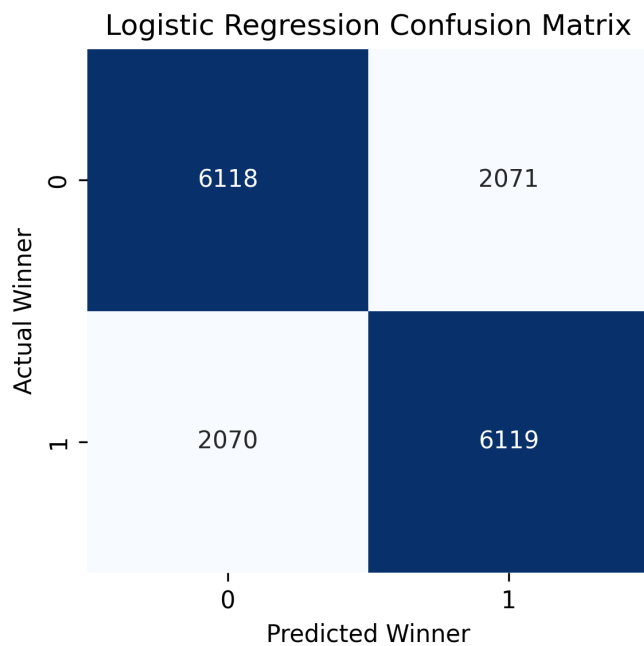


Figure 15: Logistic regression confusion matrix on the hold-out set.

6 Transformer Training Dynamics

The Transformer encoder models the per-round trajectory directly. Even without GPU acceleration it converged in under three minutes on ESTA and roughly 45 seconds on the Kaggle corpus (which is larger but contains fewer feature dimensions). Its calibration curves aligned tightly with empirical frequencies on both datasets, but the gap between ESTA and Kaggle underlines how much per-player telemetry matters: removing individual kills/damage reduces accuracy by 10 percentage points and worsens log-loss by 6x even though the architecture and optimization settings are unchanged.

7 Conclusions and Next Steps

Full parsing unlocked the entire ESTA corpus (1557 usable matches) and the 5k-match Kaggle archive. Traditional models reach $\approx 75\%$ accuracy on ESTA but drop to the high 60s on Kaggle because the latter lacks player-level stats. The Transformer crosses the 95% threshold on ESTA by consuming round-by-round context, yet it tops out at 85% on Kaggle, reinforcing the importance of richer telemetry. Future work: (i) incorporate event-level signals (utility usage, opening duel timing) for ESTA, (ii) augment Kaggle with derived features (rolling win streaks, eco detection) to reclaim performance, and (iii) run time-split validation to mimic season-wide generalization.